Summary of paper 02: carryover representation and inference for biomarker-treatment interaction tests with a predictive biomarker in aggregated N-of-1 trials under the correlation encoding of the interaction

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Problem and approach

Aggregated N-of-1 trials pool repeated within-patient on/off contrasts to validate predictive biomarkers. Drug carryover attenuates the biomarker-by-treatment interaction. The analyst must decide how to represent residual exposure, how to obtain a correctly sized test, and whether the half-life must be known. We ran an exploratory simulation study under the structure of the recommended DGP, in which the interaction is a correlation between the biomarker and the drug response and the biomarker is predictive only. The parameter values are the simulation core's own set, not the companion paper's (Table 2). An analysis ladder changes one thing per rung: the starting analysis A0 (rung 0), a binary indicator with residual AR(1) correlation A1 (rung 1), CR2 inference A2 (rung 2), an exposure-weighted predictor at the true half-life A3 (rung 3, an oracle benchmark), and the half-life chosen by AIC on the model without the interaction A4 (rung 4; G10 in the companion papers). A biomarker that is also prognostic is the subject of a companion paper (1). Supporting tables S1 to S12 are in the supplement.

Simulation design

Table 1 lists the six scenario families.

Table 2 gives the parameter values.

Table 3 defines the five analyses of the ladder, A0 to A4.

Results

The ladder at a glance

Figure 1 gives size by family.

Table 1: The six scenario families of the ladder comparisons. Every biomarker is predictive only. Within a cell every analysis is fitted to the same simulated trials; A4 was fitted to the stored trials of families 1 to 5 afterward. Family S has independent seeds for every cell. the full report describes the cells.

Family	Scenarios	Replicates
1	Base 30-cell grid: three designs, $N \in \{35, 70\}$, $c_{bm} \in \{0, 0.30, 0.45\}$, $t_{1/2} \in \{0, 0.5, 1\}$ weeks	500
2	Misspecified half-life (0.2, 2 or 4 weeks) and decay shape (Weibull $k \in \{0.5, 2\}$), analyzed as exponential	500
3	Serial correlation $\rho \in \{0.5, 0.9\}$	500
4	Patient-level dropout, completely at random or by baseline severity	500
5	Patient slope variance (SD 0.3 per week), Hybrid, $N = 70$	1,000
S	Hybrid run with independent seeds: A3 and A4, $N \in \{35, 70\}$	1,000 to 5,000

Table 2: Parameter values of the data-generating process. This paper uses the simulation core's own set ('simcore2026' in 'simulation-core.R'). The companion paper's recommended DGP uses the response and baseline values of the published code with the same correlation structure, which is described in the text. The biomarker enters the analysis centered and scaled, so its location and scale do not affect the interaction test.

Parameter	This paper	Companion paper
Natural history (TV): Gompertz max, rate, displacement	10.99, 0.42, 5	6.51, 0.35, 5
Placebo response (PB): Gompertz max, rate, displacement	6.51, 0.35, 5	6.51, 0.35, 5
Drug response (BR): Gompertz max, rate, displacement	10.99, 0.42, 5	10.99, 0.42, 5
SD of TV, PB, BR	6, 2, 8	10, 10, 8
Biomarker: mean, SD	0, 1	124.3, 15.4
Baseline severity: mean, SD	70, 10	83.1, 18.5
Within-component serial correlation ρ	0.7	0.7
Cross-component constant c_1	0.2	0.2
Biomarker-response correlation c_{bm}	0 to 0.45 by scenario	by scenario

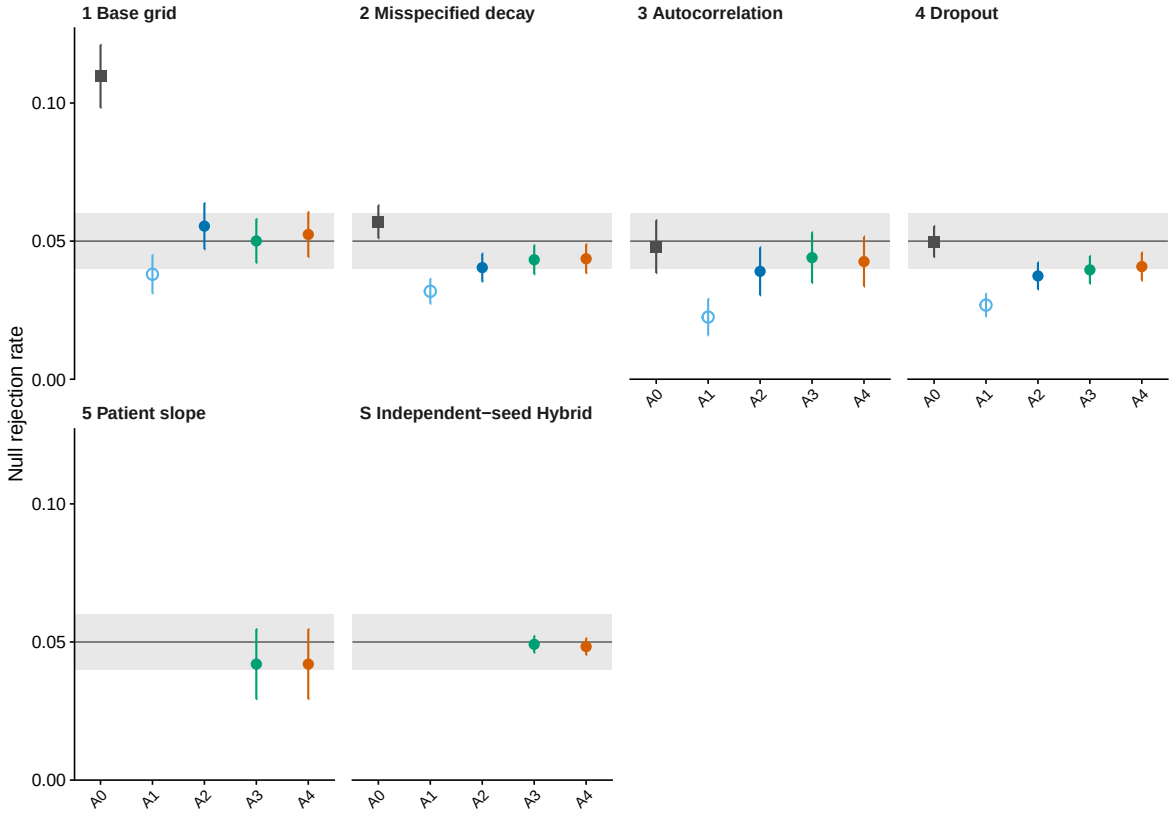


Figure 1: The size ladder. Each panel is one scenario family (Table 1). The vertical axis is the null rejection rate pooled over the null cells of the family, weighted by replicates. The horizontal axis lists the analyses in rung order, A0 to A4. Bars are 2 × the Monte Carlo standard error. The shaded band is 0.04 to 0.06, the two-standard-error band for roughly 2,000 replicates. It is a fixed tolerance around 0.05 and not a test. An analysis that was not fitted in a family has no point.

Table 3: The five analyses of the ladder. Each rung changes one thing from the one before, except the step from A0 to A1. A3 uses the true half-life and is an oracle benchmark, not a candidate analysis. The companion papers and earlier drafts use the codes in the third column; the data files use the stored codes. Sx is the symptom score, bm the biomarker, Db the binary on-drug indicator, Dbc the exposure-weighted predictor and De the design expectancy. Every analysis has a patient random intercept.

Analysis	Rung	Code in companion papers	Stored code	Formula	Inference	Assumptions
A0	0	E0	E0	Sx ~ bm + t + Db + bm:Db (+ De where expectancy varies); random intercept	lme4 with lmerTest Satterthwaite df, no residual correlation	Binary drug indicator; independent residuals given the random intercept; patients never on drug dropped
A1	1	G1	E1	Sx ~ bm + t + Db + bm:Db; random intercept, corCAR1 residuals	nlme working-covariance standard error	Binary drug indicator; AR(1) residual correlation at the stated working structure
A2	2	G6	E1cr2	Sx ~ bm + t + Db + bm:Db; random intercept, corCAR1 residuals	CR2 cluster-robust standard error, Satterthwaite df	Binary drug indicator; working covariance may be wrong
A3	3	G8	E2cr2	Sx ~ bm + t + Dbc + bm:Dbc; random intercept, corCAR1 residuals	CR2 cluster-robust standard error, Satterthwaite df	Exponential carryover at the true half-life
A4	4	G10	E7null_cr2	A3 formula; half-life chosen by ML-AIC over {0.25, 0.5, 1, 2} weeks on the model without bm:Dbc, then refit by REML with it	CR2 cluster-robust standard error, Satterthwaite df, selection ignored	Exponential carryover; true half-life near the candidate grid; selection does not use the interaction term

A4 lies in the 0.04 to 0.06 band in 6 of 6 families; A0 lies outside it in 1 of 4. Figure 2 gives power and the paired change at each step.

Table 4 gives the paired increments. In the base grid, choosing the half-life on the model without the interaction changes power against the oracle by -0.011 (paired SE 0.008).

Rungs 0 to 2: residual correlation and CR2 inference

A0 is liberal under CO and OL+BDC (Table S1), and its missing residual correlation accounts for the whole inflation (Table S2). The working-covariance test of A1 is conservative; CR2 (A2) restores nominal size (Table S3). Figures 3 and 4 give power for the five analyses on the corrected grid.

Rung 3: exposure weighting

Table 5 gives the reference cell and Figure 5 power against the half-life.

Exposure weighting recovers only part of the loss. At a one-week half-life A3 recovers 15% to 44% of A2's loss under Hybrid and 19% to 25% under OL+BDC. A3 uses the true half-life and decay form, so the rest has to be met by the design.

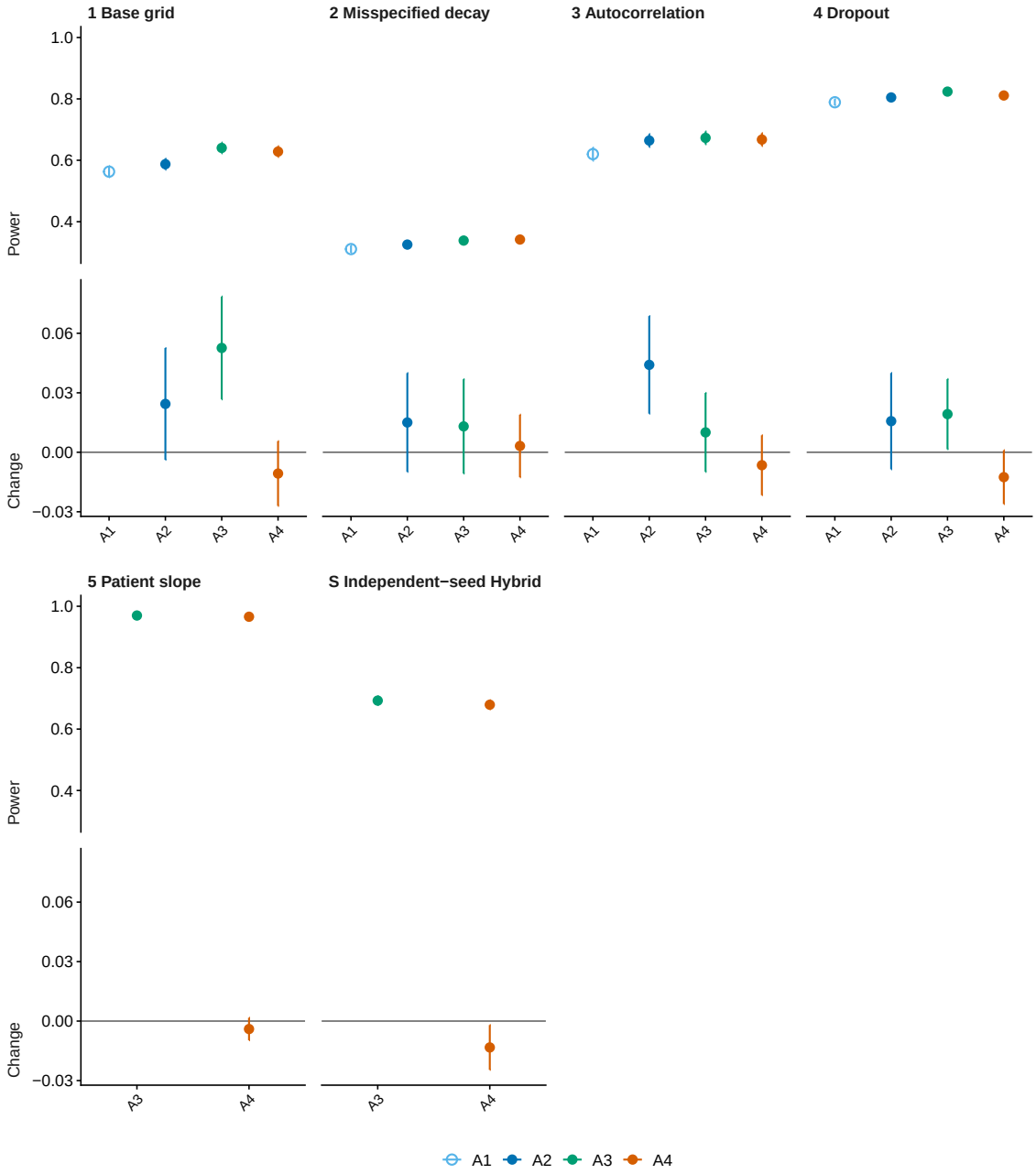


Figure 2: The power ladder. Each panel is one scenario family. A0 is omitted because its test is liberal; its size is in Figure 1. The upper part of a panel is the power at the reference alternative, pooled within the family and weighted by replicates. The lower part is the paired change in power from the preceding rung on the same trials. The horizontal axis is in rung order as in Figure 1. Each row shares one vertical scale across all families, so heights can be compared between panels. The open circle marks A1, which uses the working-covariance standard error; filled circles use CR2. Families 5 and S fitted A3 and A4 only. Bars are 2 $\times$ the Monte Carlo standard error, and the paired standard error for the changes. Paired standard errors are conservative because cells within a job share random-number streams.

Table 4: Paired increments along the ladder. Each cell is the later analysis minus the earlier one, with its paired standard error in parentheses, for size (pooled null rejection rate) and power (pooled reference alternative), by scenario family. A dash means the step has no data in that family. The standard errors are conservative because cells within a job share random-number streams. The step from A0 to A1 changes three things at once; its power change is not shown because A0 is liberal.

		Scenario family					
Step	Measure	1	2	3	4	5	S
A0 to A1	Size	-0.072 (0.013)	-0.025 (0.010)	-0.025 (0.009)	-0.023 (0.009)	–	–
	Power	–	–	–	–	–	–
A1 to A2	Size	+0.018 (0.008)	+0.009 (0.006)	+0.017 (0.007)	+0.011 (0.007)	–	–
	Power	+0.024 (0.014)	+0.015 (0.012)	+0.044 (0.012)	+0.016 (0.012)	–	–
A2 to A3	Size	-0.006 (0.007)	+0.003 (0.007)	+0.005 (0.006)	+0.002 (0.006)	–	–
	Power	+0.053 (0.013)	+0.013 (0.012)	+0.010 (0.010)	+0.019 (0.009)	–	–
A3 to A4	Size	+0.002 (0.005)	-0.000 (0.004)	-0.002 (0.004)	+0.001 (0.005)	+0.000 (0.003)	-0.001 (0.002)
	Power	-0.011 (0.008)	+0.003 (0.008)	-0.007 (0.008)	-0.013 (0.007)	-0.004 (0.003)	-0.013 (0.006)

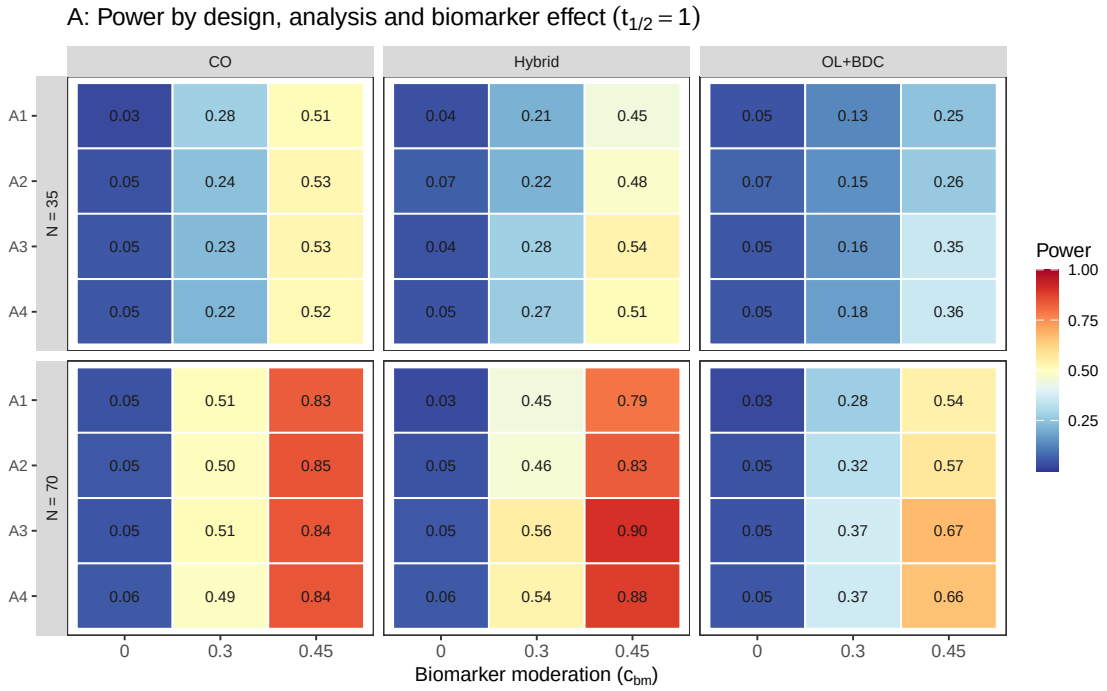


Figure 3: Power by trial design (columns), N (row blocks) and analysis (rows within block) across $c_{bm} \in \{0, 0.30, 0.45\}$ at $t_{1/2} = 1.0$ weeks, exponential DGP, corrected simulation code ($n_{sim} = 500$). The $c_{bm} = 0$ column is Type I error. A4 was fitted afterward to the same simulated trials. A0 is omitted because its test is liberal; its size is in Figure 1 and Table S1.

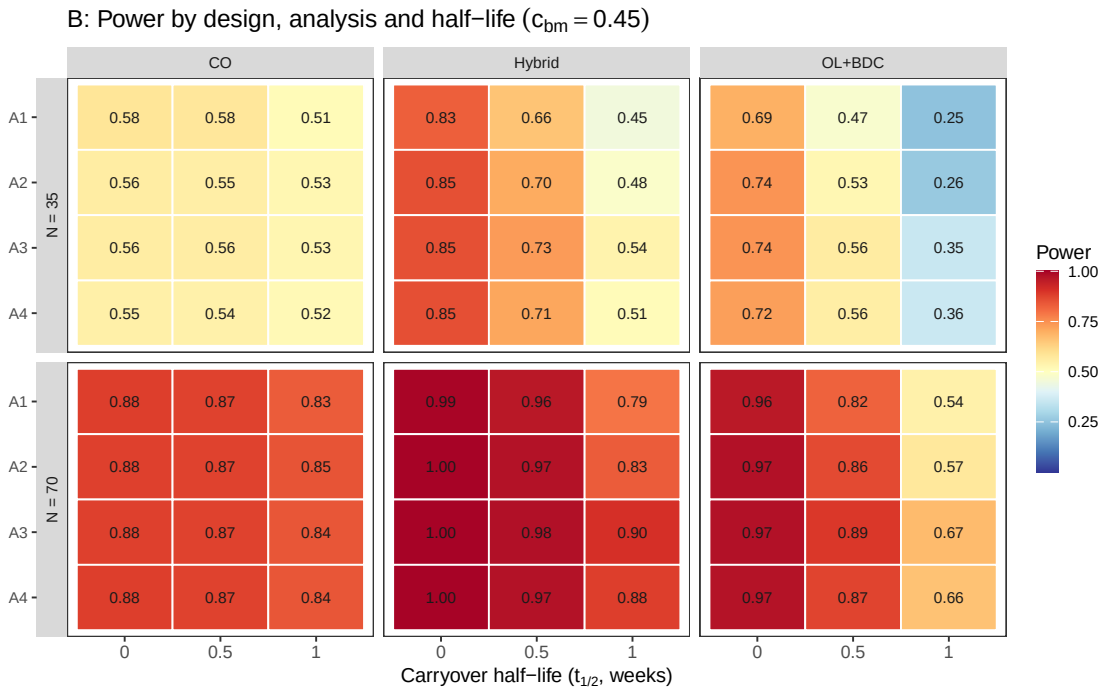


Figure 4: Power across $t_{1/2} \in \{0, 0.5, 1.0\}$ weeks at $c_{bm} = 0.45$, exponential DGP, corrected simulation code ($n_{sim} = 500$).

Table 5: Reference cell on the corrected 30-cell grid (job 27: Hybrid, $N = 70$, $c_{bm} = 0.45$, exponential decay, 500 replicates, every analysis fitted to the same trials; A4 fitted afterward to the same trials). Power is the rejection rate of $H_0 : \beta_{bm:X} = 0$ at $\alpha = 0.05$ with its Monte Carlo standard error. Type I is the rejection rate in the null cell of the same design and N. That cell exists only at $t_{1/2} = 1$ week. Empirical SD is the standard deviation of the point estimates and Mean SE the mean reported standard error. κ is their ratio. Bias is the mean estimate minus $\theta = -c_{bm} \cdot 8 = -3.6$ and is shown only for the exposure-weighted analyses A3 and A4. A1 and A2 estimate the coefficient of the binary indicator, which is a different quantity. A0 is shown for size only, because its test is liberal and its power is not comparable. A1 uses the working-covariance standard error; A2, A3 and A4 use CR2. Non-conv is the fraction of fits that did not converge or gave no p-value.

$t_{1/2}$	Analysis	Power	Type I	Empirical SD	Mean SE	κ	Bias	Non-conv
0.0	A0	–	–	–	–	–	–	0.000
	A1	0.992 (0.004)	–	0.73	0.88	1.21	–	0.000
	A2	0.996 (0.003)	–	0.73	0.71	0.98	–	0.004
	A3	0.996 (0.003)	–	0.73	0.71	0.98	-0.03	0.004
	A4	0.996 (0.003)	–	0.75	0.74	0.98	-0.16	0.000
0.5	A0	–	–	–	–	–	–	0.000
	A1	0.960 (0.009)	–	0.76	0.88	1.16	–	0.000
	A2	0.972 (0.007)	–	0.76	0.75	0.99	–	0.002
	A3	0.976 (0.007)	–	0.86	0.85	0.99	-0.02	0.006
	A4	0.972 (0.007)	–	0.86	0.81	0.94	+0.21	0.006
1.0	A0	–	0.060 (0.011)	–	–	–	–	0.000
	A1	0.790 (0.018)	0.032 (0.008)	0.79	0.89	1.12	–	0.000
	A2	0.831 (0.017)	0.046 (0.009)	0.79	0.78	0.99	–	0.004
	A3	0.904 (0.013)	0.054 (0.010)	1.05	1.05	1.00	-0.02	0.000
	A4	0.881 (0.015)	0.056 (0.010)	1.28	1.05	0.82	+0.06	0.006

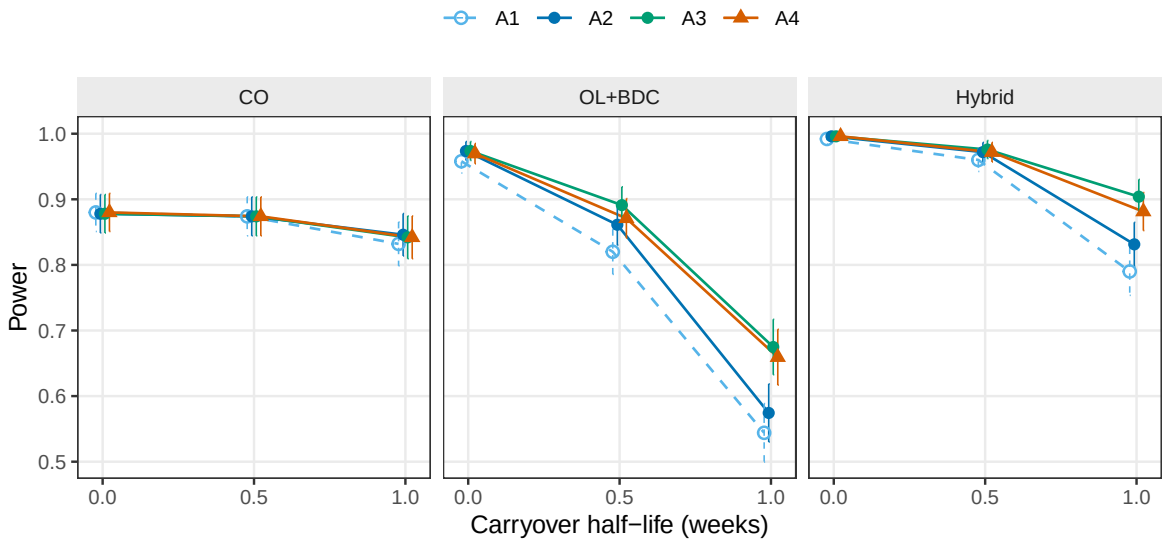


Figure 5: Power against carryover half-life by analysis and design (job 27: $N = 70$, $c_{bm} = 0.45$, exponential decay, 500 replicates). A0 is omitted because its test is liberal. The analyses are A1 (binary indicator, working-covariance standard error); A2 (binary indicator, CR2); A3 (exposure-weighted at the true half-life, CR2); and A4 (exposure-weighted at the half-life chosen by AIC without the interaction, CR2), fitted afterward to the same trials. Points at each half-life are offset horizontally for legibility. Bars are 2 × the Monte Carlo standard error.

Rung 4: choosing the half-life

Pooled by family, A4 changes power against the oracle A3 by -0.013 to +0.003 and size by -0.002 to +0.002. The selection does not recover the half-life (Table S4). In the Hybrid run with independent seeds A4 differs from A3 under the null by -0.0012 to -0.0002 (Tables S5 and S6).

Stress scenarios

The misspecification cells and scenario grids are in Tables S7 to S10. Table 6 imposes monotone dropout.

The A4 Type I error under dropout is 0.026 to 0.054. The heavy-missingness case and the patient-specific linear trend are in Tables S11 and S12.

Calibration and coverage

Figure 6 gives the calibration ratio and Table 7 the coverage of A3 and A4.

Table 6: Type I error and power under monotone dropout (Hybrid, $t_{1/2} = 0.5$ week; the rate is the fraction of patients with an incomplete series; $n_{sim} = 500$; band $[0.031, 0.069]$). A4 was fitted afterward to the same trials. A0 has no power column because its test is liberal.

	Type I error					Power, $c_{bm} = 0.45$			
Dropout (rate, N)	A0	A1	A2	A3	A4	A1	A2	A3	A4
MAR 0.1, $N = 35$	0.040	0.028	0.038	0.042	0.044	0.672	0.683	0.719	0.697
MAR 0.2, $N = 35$	0.052	0.034	0.042	0.046	0.046	0.638	0.651	0.687	0.669
MAR 0.3, $N = 35$	0.054	0.030	0.046	0.044	0.048	0.596	0.593	0.630	0.593
MCAR 0.1, $N = 35$	0.044	0.030	0.042	0.048	0.054	0.674	0.697	0.716	0.700
MCAR 0.2, $N = 35$	0.046	0.024	0.038	0.040	0.042	0.644	0.663	0.686	0.663
MCAR 0.3, $N = 35$	0.044	0.024	0.034	0.036	0.036	0.608	0.620	0.646	0.623
MAR 0.1, $N = 70$	0.058	0.026	0.038	0.040	0.038	0.956	0.970	0.980	0.974
MAR 0.2, $N = 70$	0.056	0.022	0.038	0.038	0.040	0.936	0.964	0.972	0.974
MAR 0.3, $N = 70$	0.056	0.038	0.038	0.040	0.038	0.910	0.940	0.940	0.946
MCAR 0.1, $N = 70$	0.056	0.022	0.032	0.036	0.038	0.958	0.970	0.978	0.978
MCAR 0.2, $N = 70$	0.052	0.020	0.030	0.032	0.026	0.950	0.964	0.976	0.970
MCAR 0.3, $N = 70$	0.040	0.024	0.032	0.032	0.038	0.926	0.942	0.956	0.946

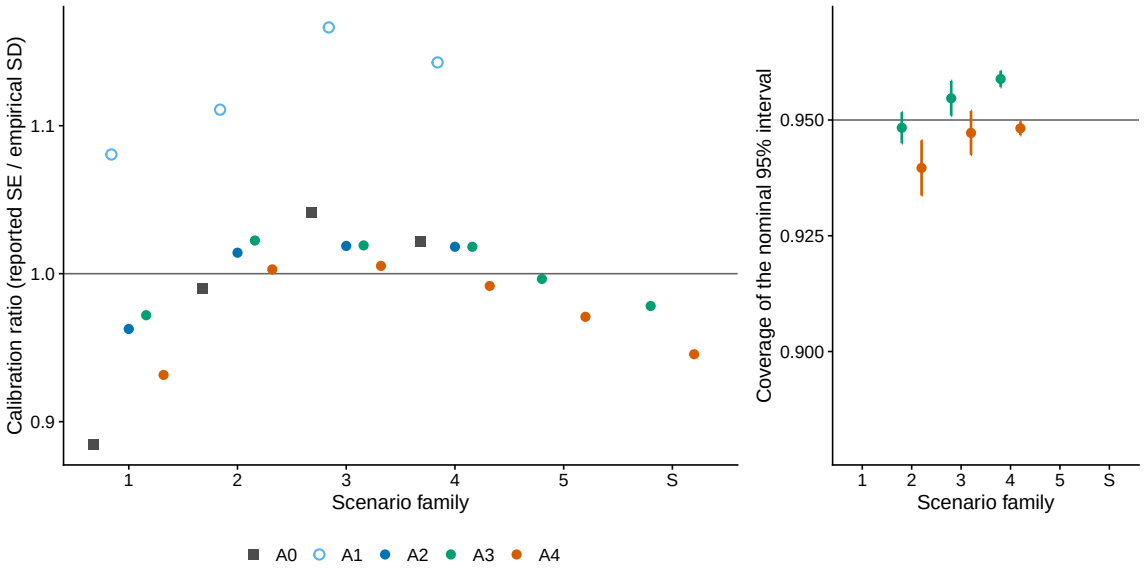


Figure 6: Calibration and coverage. Left: the calibration ratio, the mean reported standard error divided by the empirical standard deviation of the estimates, averaged without weights over the cells of each scenario family, by analysis. A value of one is calibrated and a value above one is conservative. Right: the coverage of the nominal 95 percent interval of A3 and A4 in families 2 to 4, as the mean of the cell lower bounds and upper bounds. The bars span the two bounds. The coverage bounds are not Monte Carlo intervals. Families 1, 5 and S have no coverage output.

Table 7: Coverage of the nominal 95 percent CR2 interval for A3 and A4 over the predictive-only alternative cells of the supplementary study without heavy missingness, on the calibrated scale ($\theta = -c_{bm}\sigma_{BR}$). The Satterthwaite degrees of freedom are not stored per replicate and were recovered from the stored t and p where $|t| \geq 0.5$; the remaining replicates are classified as covering, not covering, or ambiguous, which gives the lower and upper bounds. A3 from '34-referee-reanalysis.R', A4 from '43-g10-coverage.R'.

Method	N	Cells	Mean coverage (lower, upper)	Minimum (lower)
A3	35	15	0.951, 0.958	0.938
A3	70	15	0.951, 0.954	0.915
A4	35	15	0.946, 0.955	0.914
A4	70	15	0.934, 0.941	0.869

Table 8: Scoreboard of the analyses over the scenario families. Families in band counts the families whose pooled null rejection rate lies in 0.04 to 0.06, out of the families in which the analysis was fitted; it is descriptive and does not separate the analyses (Table 9 compares size with the oracle directly). Smallest and largest size are cell-level null rejection rates. Power is the mean over families of the pooled power as a percent of A4 in the same family. The worst power loss is against A4 in points of power, by family and by cell; a negative value is a gain. Coverage is the mean of the nominal 95 percent interval over families with coverage output (A3 and A4 only). A dash means not available; the power columns of A0 are dashed because its test is liberal. A4 is the reference, so its losses are zero.

Analysis	Families in band	Smallest size	Largest size (cell)	Power, percent of A4	Worst family loss, points (family)	Worst cell loss, points (cell)	Coverage
A0	3 of 4	0.034	0.150 (CO, $N = 35$, $c_{bm} = 0.00$, $t_{1/2} = 1$)	–	–	–	–
A1	0 of 4	0.016	0.048 (OL+BDC, $N = 35$, $c_{bm} = 0.00$, $t_{1/2} = 1$)	92.7	6.5 (1)	12.1 ($\rho = 0.9$, $N = 35$, $c_{bm} = 0.30$, $t_{1/2} = 0.5$)	–
A2	2 of 4	0.030	0.074 (OL+BDC, $N = 35$, $c_{bm} = 0.00$, $t_{1/2} = 1$)	96.9	4.1 (1)	9.3 (OL+BDC, $N = 35$, $c_{bm} = 0.45$, $t_{1/2} = 1$)	–
A3	5 of 6	0.032	0.054 (Hybrid, $N = 70$, $c_{bm} = 0.00$, $t_{1/2} = 1$)	100.9	0.3 (2)	2.0 (OL+BDC, $N = 35$, $c_{bm} = 0.30$, $t_{1/2} = 1$)	0.954
A4	6 of 6	0.026	0.056 (Hybrid, $N = 70$, $c_{bm} = 0.00$, $t_{1/2} = 1$)	100.0	0	0.0 ($t_{1/2} = 0.2$ wk, $N = 35$, $c_{bm} = 0.25$, $t_{1/2} = 0.2$)	0.945

Which analysis to use

Table 8 scores the analyses over the families.

The band counts do not separate the analyses. Table 9 compares the size of A4 with the oracle's directly.

Sample size for A4

Table 10 gives the total N for 80 percent power of A4, and Figure 7 the power curves under this paper's parameter set. At a 1-week half-life the binary indicator (A2) needs 15 to 18 percent more patients than A4 in the Hybrid design. Under the starting values of the companion paper the required N is roughly twice as large, and it depends strongly on the serial correlation

Table 9: Size of A4 against the oracle A3, by family: the paired difference in null rejection rate on the same trials, its paired standard error, and the 90 percent interval. An interval inside ± 0.01 is equivalence by two one-sided tests at the 5 percent level. The paired SE is the replicate-weighted mean of the cell SEs, an upper bound, so the interval is conservative. The margin was chosen after the results were seen. Family S has independent seeds per cell; the cells of families 2 to 4 share random-number streams.

Family	A4 − A3	Paired SE	90% interval	Within ±0.01
S (independent seeds)	-0.0008	0.0016	-0.0033 to +0.0018	yes
1	+0.0023	0.0047	-0.0054 to +0.0101	no
2	+0.0000	0.0043	-0.0071 to +0.0071	yes
3	-0.0015	0.0041	-0.0083 to +0.0053	yes
4	+0.0012	0.0046	-0.0064 to +0.0088	yes
5 (one cell)	+0.0000	0.0028	-0.0047 to +0.0047	yes

Table 10: Total N for 80 percent power of A4 at $\alpha = 0.05$, with a 95 percent Monte Carlo interval, by design, carryover half-life (weeks) and c_{bm}, under the parameter set of this paper and under the starting values of the companion paper. From a probit curve in $c_{bm}\sqrt{N}$ fitted to five values of N per c_{bm}, 500 replicates each; the interval does not include the approximation of the curve model (about 6 percent). $\rho = 0.7$, exponential decay, complete data, predictive-only biomarker.

		This paper, c_{bm}			Starting values, c_{bm}		
Design	$t_{1/2}$	0.2	0.3	0.4	0.2	0.3	0.4
CO	0.0	302 (295–310)	135 (131–138)	76 (74–78)	566 (551–582)	252 (245–259)	142 (138–146)
	0.5	302 (295–311)	135 (131–138)	76 (74–78)	571 (556–587)	254 (247–261)	143 (139–147)
	1.0	316 (308–324)	141 (137–144)	79 (77–81)	614 (598–631)	273 (266–281)	154 (150–158)
OL+BDC	0.0	198 (193–203)	88 (86–91)	50 (49–51)	418 (407–431)	186 (181–192)	105 (102–108)
	0.5	292 (285–300)	130 (127–134)	73 (72–75)	612 (595–630)	272 (265–280)	153 (149–158)
	1.0	474 (459–490)	211 (204–218)	119 (115–123)	1010 (975–1048)	449 (434–466)	253 (244–262)
Hybrid	0.0	151 (146–156)	67 (65–70)	38 (37–39)	310 (302–320)	138 (134–142)	78 (76–80)
	0.5	203 (198–209)	91 (88–93)	51 (50–53)	420 (408–432)	187 (182–192)	105 (102–108)
	1.0	292 (284–299)	130 (127–133)	73 (71–75)	613 (597–631)	273 (266–281)	154 (150–158)

(Table S13 of the supplement). Table S14 gives the N for A2 in every cell. At the sample size of the companion paper's closed-form paired-difference rule, A4 had power 0.844 to 0.936 under the starting values wherever that sample size was at most 400 (Table S15), so the rule is conservative for A4.

Recommendation

For a Hybrid trial with a biomarker known to be predictive only, at $N \leq 70$, use A4: the exposure-weighted predictor with the half-life chosen by AIC on the model without the interaction, refitted by REML with the interaction and tested with CR2. Its size is equivalent to the oracle's within ± 0.01 in the run with independent seeds and in 4 of the other five families. In the Hybrid cells it gains 1.3 points of power over A2 on average, up to 7.1 at long half-lives, and is behind by at most 1.0 point in 6 cells.

Whether a biomarker is also prognostic is rarely known before a trial. For a biomarker of unknown role the default is the analysis of the companion paper (1), which shares A4's

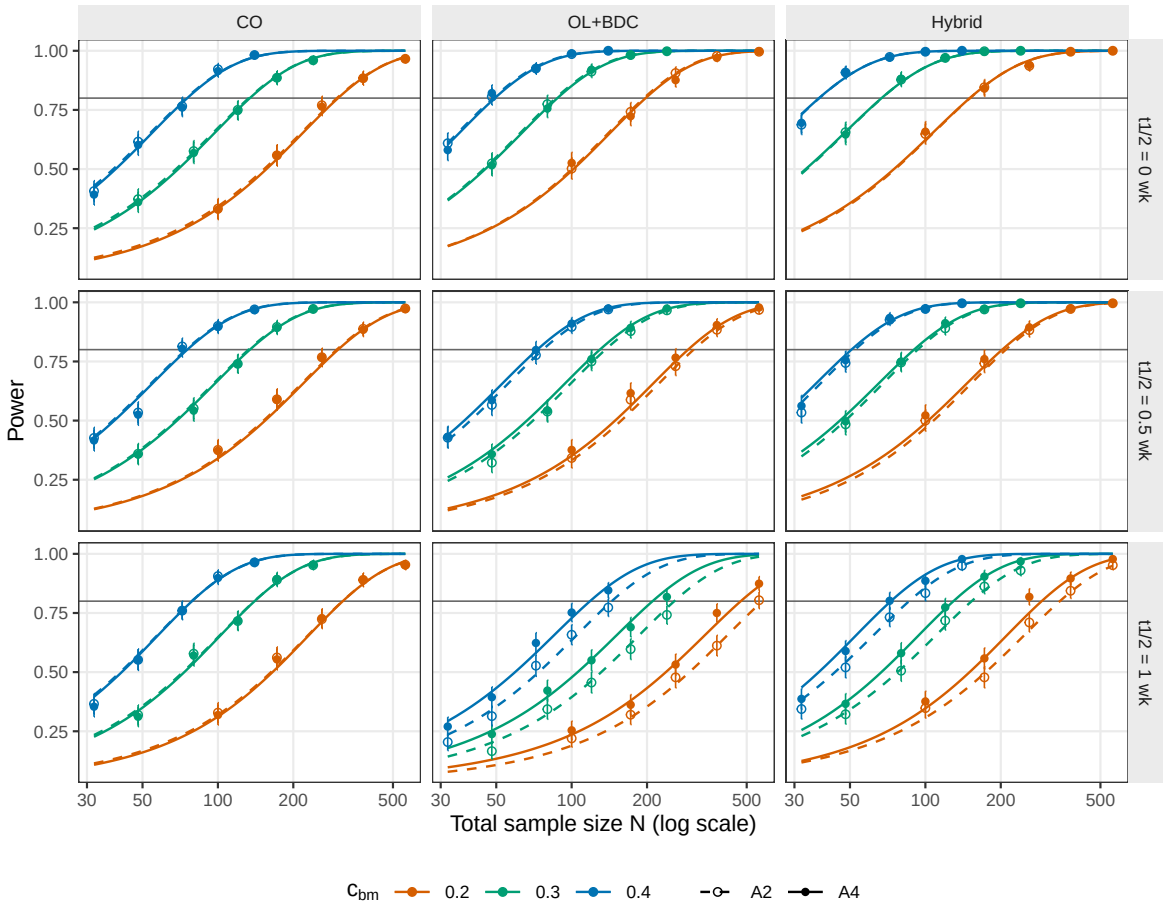


Figure 7: Power of A4 (filled points, solid lines) and A2 (open points, dashed lines) against total N under the parameter set of this paper, by design (columns) and carryover half-life (rows), for $c_{bm} \in \{0.2, 0.3, 0.4\}$. Points are simulated rejection rates (500 replicates) with $2 \times$ the Monte Carlo standard error; lines are the fitted probit curves in $c_{bm}\sqrt{N}$. The horizontal line is 80 percent power. The same figure under the starting values is Figure S1.

exposure-weighted predictor, half-life rule and CR2 test. The recommendation is conditional on the correlation encoding, on one parameter set and on one design.

References

1. pmsimstats team. Prognostic biomarkers and the biomarker-treatment interaction test in aggregated N-of-1 trials. 2026.